Francesco Dorati

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Research interests: Physical AI · Computer Vision · Reinforcement Learning · Autonomous Systems

Education

Politecnico di Milano – *MSc in Computer Science and Engineering* — Milan, Italy
Specialization in Artificial Intelligence — *2025 - Present*

- **Beijing Institute of Technology** – *Summer Program* — Zhuhai, China
  Intelligent Robotics Engineering — *Jul 2026*
- **UPC Barcelona** – *UNITECH Exchange* — Barcelona, Spain
  Courses: *Computer Vision, Machine Learning, NLP, Advanced Processor Architecture* — *Feb – Jun 2026*

Politecnico di Milano – *BSc in Computing Systems Engineering* — Milan, Italy
Final Grade: *100/110* — *2022 - 2025*
Final Projects (Avg. Score: 98.6%): *Algorithms & Data Structures (C), Software Engineering (Java), FPGA Design (VHDL)*

Publications

Foundation ViTs Can Learn Where to Act: Task Affordance Segmentation and Surface Normals
Francesco Dorati, Tommaso F. Banfi
ECCV 2026 Workshop, under review

Bayesian Optimization for Learning Nonlinear MPC in Autonomous Agent Navigation
Lorenzo Ortolani, Gabriel Voss, Gabriele Beltrami, **Francesco Dorati**, Tommaso F. Banfi
IEEE ICRA 2026 Workshop

Optimizing Initial Path Finding in Informed-RRT* with a Novel Map-Adaptive Sampling Technique
Tommaso F. Banfi, **Francesco Dorati**, Nicola Manzoni, Jesús Martínez-Gómez
7th Iberian Robotics Conference (ROBOT 2024), IEEE Xplore

Experience & Research

AI Researcher — Oct 2025 – Present
Talos Robotics AI — Early-stage humanoid robotics startup — *Milan, Italy*
Contributing to a platform for training, data collection, and deployment of VLA manipulation policies on humanoid robots for industrial environments; team secured **25k industrial research contract** (VHIT) and **10k award** (TEF Ignition Program).

- Developing a **multi-task perception pipeline** for zero-shot robotic affordance detection: predicts 7-class pixel-level affordance masks and surface normals from RGB using **DINOv2** with multi-scale feature fusion and RGB skip connections
- **80.3% mean-IoU** on the official UMD benchmark, **surpassing prior SOTA** (Mask2Former, 75.9%)

R&D: Hierarchical Navigation Stack for Industrial AGV — Jan 2026 – Present
DevNut (Italian tech startup) — *Remote*
Sensor-based AGV navigation in **Webots**, replacing fixed floor-line infrastructure

- Designed a **two-layer planner**: global visibility-graph + A* for path planning using local occupancy-grid planner from LiDAR
- Designed a **map-differencing** approach separating static walls from dynamic obstacles, with per-frame cluster extraction and velocity-aware tracking

Undergraduate Researcher – FALCO Autonomous Drone — Nov 2023 – Feb 2025
Automation Engineering Association, Politecnico di Milano — *Milan, Italy*
Developed novel **path planning algorithm** for autonomous drone navigation in 3D environments

- Achieved **85% faster** initial path discovery and **78% reduction in node count** than state-of-the-art Informed-RRT*
- Proposed a **map-adaptive probabilistic sampling strategy** and implemented the full algorithm in **C++**, with a **Python** benchmarking and visualization framework

Awards & Fellowships

UNITECH International Programme – cohort 2025 — *2025 – 2026*
Competitive international programme combining academic exchange and industry internship

Tiny_Hack Competition – 3rd place — *2025*
Embedded computer vision on microcontrollers

Best Freshmen Award – Politecnico di Milano — *2023*

Technical Skills

- **Programming:** Python, C++, C, Java, MATLAB, SQL
- **ML & Deep Learning:** PyTorch, Scikit-learn, Model Quantization, DINOv2
- **Computer Vision:** OpenCV, YOLO, Image Segmentation, Object Tracking, Sensor Fusion
- **Robotics & RL:** ROS/ROS2, PPO, Actor-Critic, Imitation Learning, Isaac Sim, MuJoCo, Gazebo
- **Hardware & Embedded:** VHDL/FPGA, Raspberry Pi, Arduino
- **Infrastructure:** Linux, Docker, Git, Edge Deployment

Courses & Certifications

Machine Learning Specialization – Stanford University, DeepLearning.AI [link] *2025*
Deep Learning Specialization – DeepLearning.AI [link] *2024*
CS50x: Introduction to Computer Science – Harvard University [link] *2021*

Languages

English – IELTS Academic 7.5; Cambridge IGCSE grade A *Fluent*
Italian *Native*